

Sami Melhem

512-713-9742 | SaMiLMelhem23@gmail.com | samimelhem.com/ | linkedin.com/sami-melhem | github.com/SamiMelhem

EDUCATION

Texas A&M University

Master of Computer Science

- **GPA: 4.0.**

May 2027

College Station, TX

Texas A&M University

Bachelor of Science in Computer Science, Cybersecurity Minor

- **GPA: 3.8.**

May 2026

College Station, TX

- **Coursework:** Software Testing & Quality Assurance, Database Systems, Artificial Intelligence, Software Engineering.

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, HTML, CSS, Go, Rust, R

Frameworks: React.js, Node.js, Express.js, Flask, Dash, Django, Spring Boot, Next.js

Data Science: Pandas, Scikit-learn, TensorFlow, PyTorch, NumPy, Matplotlib, Plotly, OpenCV, Excel

AI/ML: RAG, Object-Oriented Programming, Model Optimization, NLP, Multi-Agent Systems, LLM Architecture

Tools: Git, Docker, Selenium, Pytest, Postman, Electron, Jira, Tiktoken, CLI, Vercel, Power BI, Tableau, RESTful APIs, Cursor

Databases: MySQL, PostgreSQL, NoSQL, AWS, GCP, Azure, Salesforce, ChromaDB, Linux, IPFS, Cloud Infrastructure

Methodologies: Agile, Scrum, Test-Driven Development, Behavior-Driven Development, SDLC, CI/CD, Automated Testing

EXPERIENCE

Momentum Technologies Inc.

Machine Learning Engineer Intern

June 2025 – Nov 2025

Carrollton, TX

- Developed mathematical and ML models that cut error rates by **30%**, strengthening overall product reliability.
- Accelerated development efficiency by decreasing coding time by **25%** through designing forecasting models.
- Optimized project turnaround speed to elevate system uptime to **99.5%** through accelerating development efficiency.

INTERA Incorporated

Data Science Intern

May 2024 – August 2024

Austin, TX

- Architected ML-driven data pipelines to boost climate-change prediction accuracy by **30%**.
- Streamlined data and model workflows to slash processing time by **40%**, improving computational throughput.
- Integrated reinforcement-learning and optimization pipelines to achieve **95%** model accuracy.

Pivotal Research Inc.

Data Science Intern

May 2024 – July 2024

Edmonton, Alberta

- Engineered scalable infrastructure handling **1M+** daily transactions, elevating pipeline performance by **25%**.
- Collaborated with cross-functional teams to build advanced ML models, raising predictive accuracy by **30%**.
- Conducted targeted Python/SQL analyses, trimming analysis time by **35%** and expediting decision cycles.

PROJECTS

AI-Powered Data Breach Hub

August 2025 – December 2025

- Architected a data intelligence platform aggregating **3.1k+** incidents per year, enabling real-time sector and threat analysis.
- Deployed GenAI pipelines detecting **+42%** ransomware growth in universities and data-theft disclosures in hospitals.
- Provisioned a scalable, privacy-safe AWS architecture ensuring **100%** PII-free ingestion and high-throughput analytics.

Notion RAG CLI Tool

July 2025 – July 2025

- Compiled cost-effective AI Notion knowledge retrieval by developing a RAG system in Python/Gemini.
- Achieved **1.4s** document search response time by implementing ChromaDB with embeddings and semantic search.
- Built token-tracking and API-monitoring stream to safeguard production reliability and reduce operating costs.

SaveGas

April 2025 – May 2025

- Saved users an average of **\$0.15-\$0.30/gal** by developing a real-time gas price tracking app with **100+** gas stations.
- Implemented location-based authentication to secure user data across **100+** stations and pricing requests.
- Tuned API cooldown parameters to ensure seamless, cross-platform price updates and UX consistency.

Financial Modeling Tool

June 2024 – August 2024

- Developed a Python-based modeling tool that delivered **80%+** accuracy on stock and revenue forecasts.
- Engineered an object-oriented backend to support high-volume data ingestion, enabling scalable analysis.
- Designed interactive Dash/Matplotlib/Plotly dashboards to empower analysts with real-time insights.

LEADERSHIP

Carnival of Animals – Vice President

January 2024 – Present

- Managed a **\$2500 budget** for **50+** performances at church and wedding venues, ensuring seamless event execution.
- Organized weekly jam sessions for **60** members, exploring over **100** orchestral compositions, fostering musical development.
- Coordinated events and client communications to tailor musical programs for audiences, helping ensemble reputation.